

Plastics Deformation of Metals and Alloys

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Contents

- **Plastics Deformation of Metals and Alloys:**
 - Internal stresses.
 - Role of dislocation.
 - Slip.
 - Frank-Read source.
 - Twinning.
 - Strain hardening.

Internal stresses:

Internal stresses are stresses that occur within a material without any external force. They can be caused by several factors, including.

- **Mechanical processing:**

Processes such as forging, rolling, and extrusion can cause localized deformation.

- **Heat treatments**

Rapid heating and cooling during heat treatment processes can introduce temperature gradients.

- **Welding**

Uneven heating and cooling during welding can cause localized thermal expansion and contraction.

- **Phase transformations**

Changes in the material phase structure during heat treatment or cooling can lead to residual stresses.

The five types of stress that can occur in metals are: shear stress, bending stress, torsion stress, tensile stress, and compressive stress

Why Deformation Studies

- With a knowledge of the nature of dislocations and the role they play in the plastic deformation process,
- We are able to understand the underlying mechanisms of the techniques that are used to strengthen and harden metals and their alloys;
- Thus, it becomes possible to design and tailor the mechanical properties of materials
- for example, the strength or toughness of a metal-matrix composite.
- Also, understanding the mechanisms by which polymers elastically and plastically deform allows one to alter and control their moduli of elasticity and strengths

DEFORMATION MECHANISMS FOR METALS

- Metallic materials may experience two kinds of deformation:

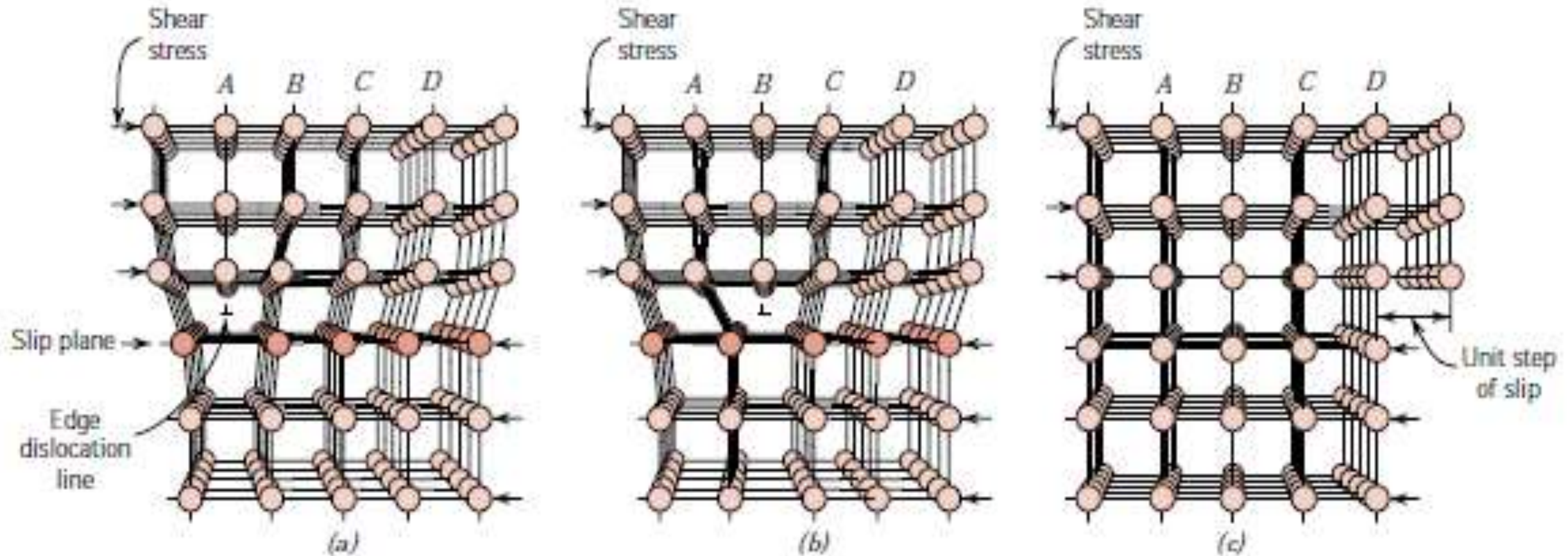
Elastic and plastic.

- Plastic deformation is permanent, and strength and hardness are measures of a material's resistance to this deformation.
- On a microscopic scale, plastic deformation corresponds to the net movement of large numbers of atoms in response to applied stress.
- During this process, interatomic bonds must be ruptured and then reformed. Furthermore, plastic deformation most often involves the motion of dislocations, linear crystalline defects

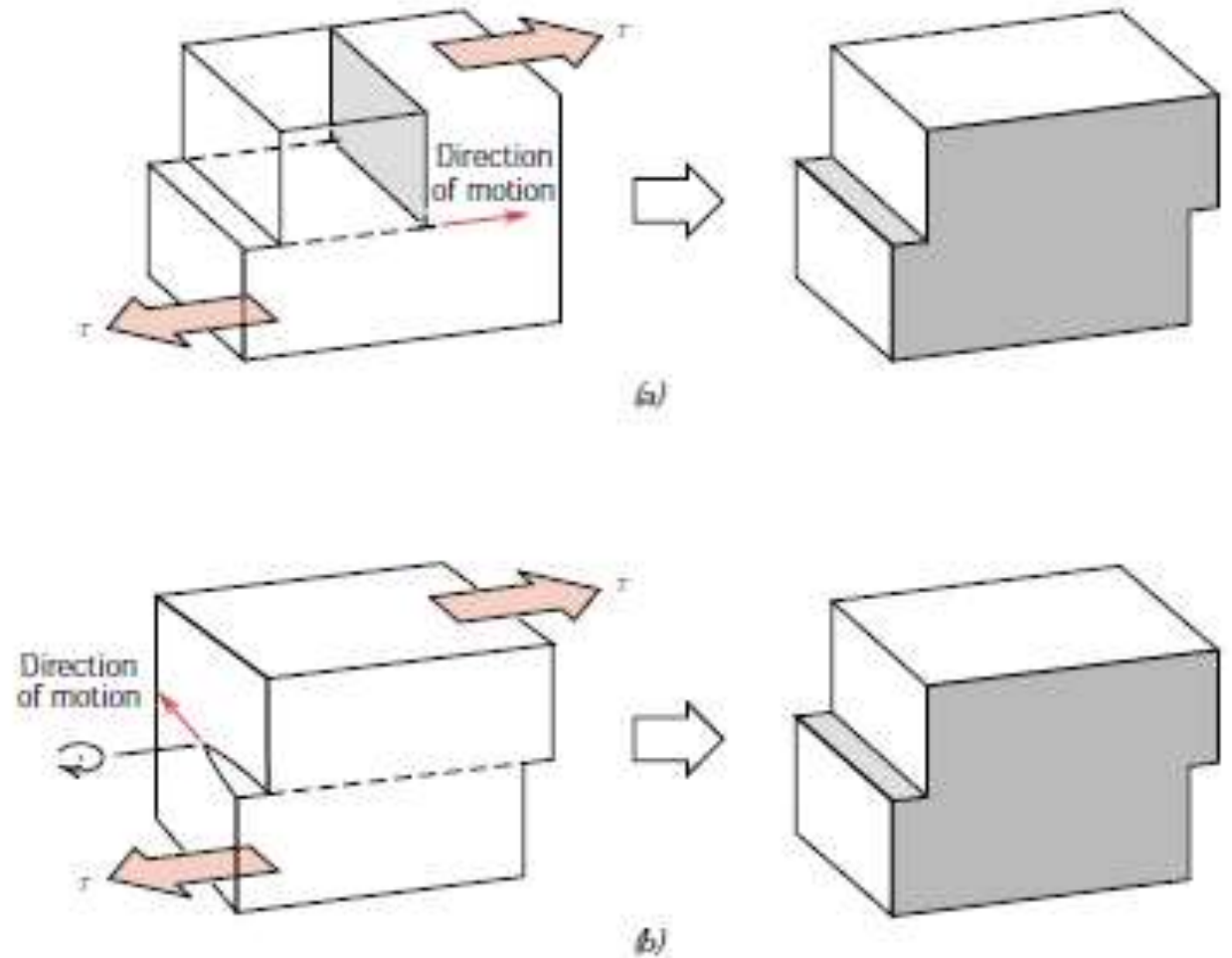
BASIC CONCEPTS OF DISLOCATIONS

- Edge and screw are the two fundamental dislocation types. In an edge dislocation, localized lattice distortion exists along the end of an extra half-plane of atoms, which also defines the dislocation line (Figure).
- A screw dislocation may be thought of as resulting from shear distortion; its dislocation line passes through the center of a spiral, atomic plane ramp (Figure).
- Many dislocations in crystalline materials have both edge and screw components; these are mixed dislocations (Figure).

- Atomic rearrangements that accompany the motion of an edge dislocation as it moves in response to an applied shear stress. (a) The extra halfplane of atoms is labeled A. (b) The dislocation moves one atomic distance to the right as A links up to the lower portion of plane B; in the process, the upper portion of B becomes the extra half-plane. (c) A step forms on the surface of the crystal as the extra half-plane exits.



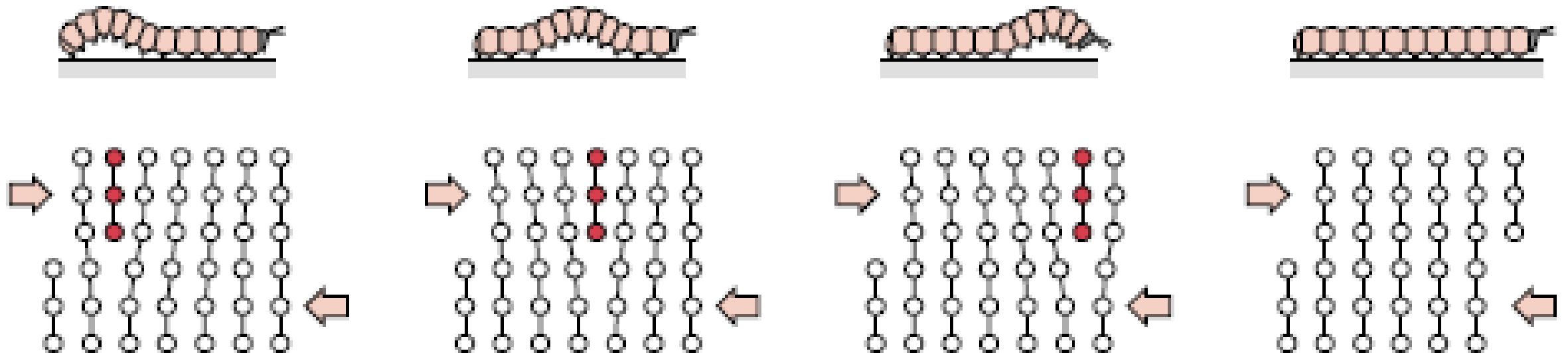
- The formation of a step on the surface of a crystal by the motion of (a) an edge dislocation and (b) a screw dislocation.
- Note that for an edge, the dislocation line moves in the direction of the applied shear stress ;
- for a screw, the dislocation line motion is perpendicular to the stress direction.



Slip:

- The process by which plastic deformation is produced by dislocation motion is termed **slip**;
- the crystallographic plane along which the dislocation line traverses is the *slip plane*, as indicated in Figure 1.
- Macroscopic plastic deformation simply corresponds to permanent deformation that results from the movement of dislocations, or slip, in response to an applied shear stress, as represented in Figure 2*a*.

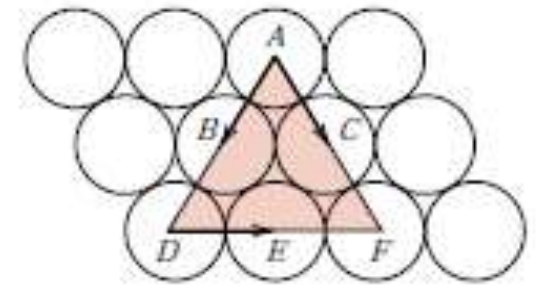
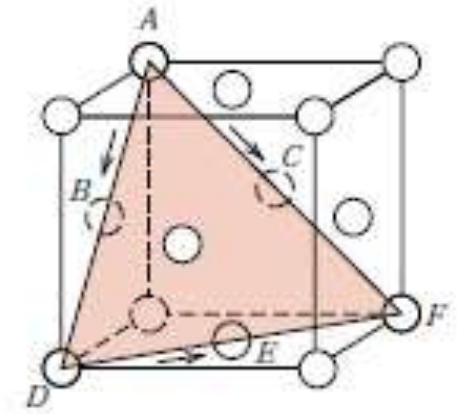
- Dislocation motion is analogous to the mode of locomotion employed by a caterpillar (Figure 3 below).
- The caterpillar forms a hump near its posterior end by pulling in its last pair of legs a unit leg distance. The hump is propelled forward by repeated lifting and shifting of leg pairs.
- When the hump reaches the anterior end, the entire caterpillar has moved forward by the leg separation distance.
- The caterpillar hump and its motion correspond to the extra half-plane of atoms in the dislocation model of plastic deformation.



SLIP SYSTEMS

- Dislocations do not move with the same degree of ease on all crystallographic planes of atoms and in all crystallographic directions.
- Ordinarily there is a preferred plane, and in that plane there are specific directions along which dislocation motion occurs. This plane is called the *slip plane*;
- it follows that the direction of movement is called the *slip direction*. This combination of the slip plane and the slip direction is termed the **slip system**.
- The slip system depends on the crystal structure of the metal and is such that the atomic distortion that accompanies the motion of a dislocation is a minimum.
- For a particular crystal structure, the slip plane is that plane having the most dense atomic packing, that is, has the greatest planar density.
- The slip direction corresponds to the direction, in this plane, that is most closely packed with atoms, that is, has the highest linear density. Planar and linear atomic densities.

- Consider, for example, the FCC crystal structure, a unit cell of which is shown in Figure *a*. There is a set of planes, the 111 family, all of which are closely packed. A (111)-type plane is indicated in the unit cell;
- in Figure *b*, this plane is positioned within the plane of the page, in which atoms are now represented as touching nearest neighbors.
- Slip occurs along 110-type directions within the 111 planes, as indicated by arrows in Figure . Hence, 111110 represents the slip plane and direction combination, or the slip system for FCC.
- Figure *b* demonstrates that a given slip plane may contain more than a single slip direction. Thus, several slip systems may exist for a particular crystal structure; the number of independent slip systems represents the different possible combinations of slip planes and directions.
- For example, for face-centered cubic, there are 12 slip systems: four unique 111 planes and, within each plane, three independent 110 directions.



(b) The {111}{110} slip system shown within an FCC unit cell. (b) The (111) plane from (a) and three 110 slip directions (as indicated by arrows) within that plane comprise possible slip systems.

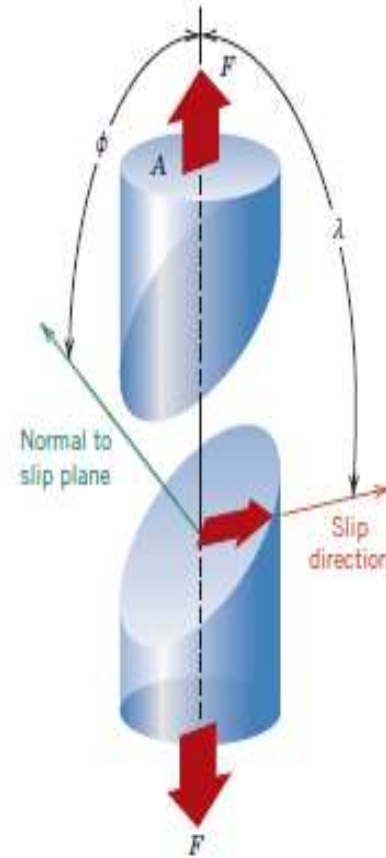
The possible slip systems for BCC and HCP

- The possible slip systems for BCC and HCP crystal structures are listed in Table . For each of these structures, slip is possible on more than one family of planes (e.g., 110, 211, and 321 for BCC).
- For metals having these two crystal structures, some slip systems are often operable only at elevated temperatures.
- Metals with FCC or BCC crystal structures have a relatively large number of slip systems (at least 12). These metals are quite ductile because extensive plastic deformation is normally possible along the various systems.
- Conversely, HCP metals, having few active slip systems, are normally quite brittle.

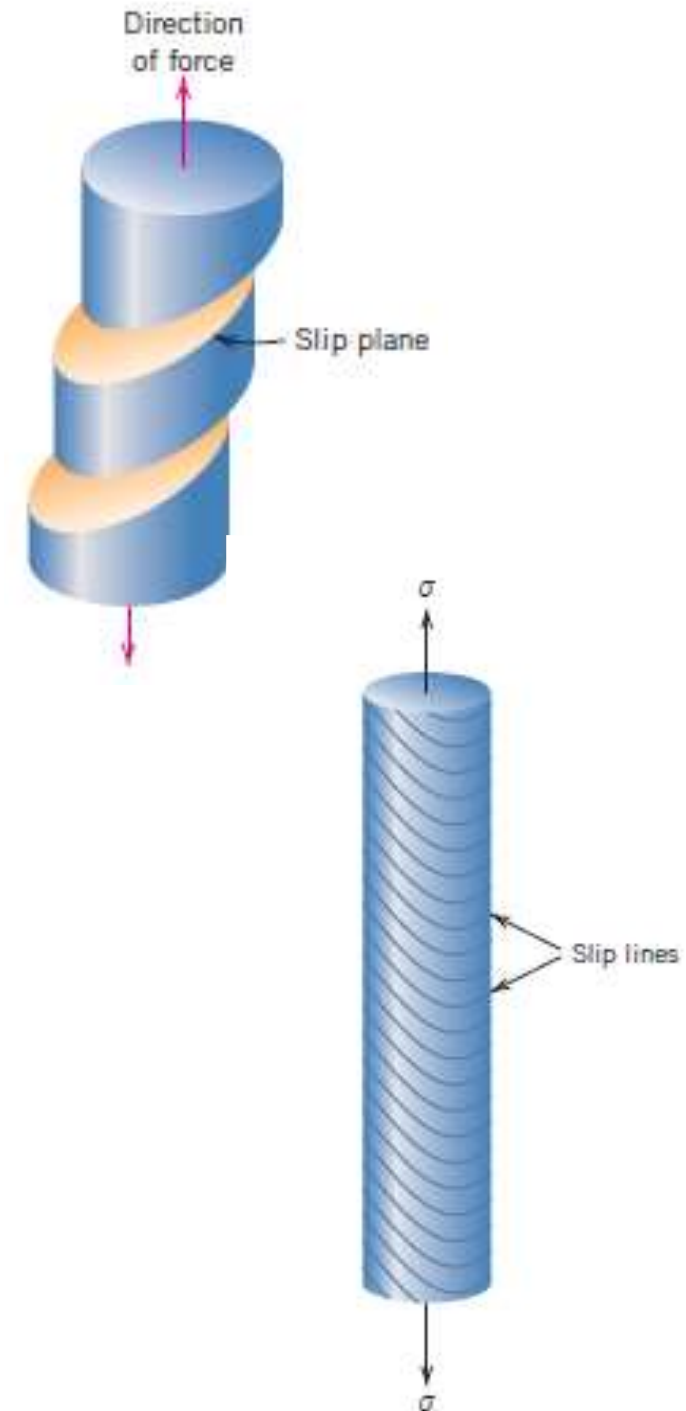
<i>Metals</i>	<i>Slip Plane</i>	<i>Slip Direction</i>	<i>Number of Slip Systems</i>
Face-Centered Cubic			
Cu, Al, Ni, Ag, Au	{111}	$\langle 1\bar{1}0 \rangle$	12
Body-Centered Cubic			
α -Fe, W, Mo	{110}	$\langle \bar{1}11 \rangle$	12
α -Fe, W	{211}	$\langle \bar{1}11 \rangle$	12
α -Fe, K	{321}	$\langle \bar{1}11 \rangle$	24
Hexagonal Close-Packed			
Cd, Zn, Mg, Ti, Be	{0001}	$\langle 11\bar{2}0 \rangle$	3
Ti, Mg, Zr	{10 $\bar{1}$ 0}	$\langle 11\bar{2}0 \rangle$	3
Ti, Mg	{10 $\bar{1}$ 1}	$\langle 11\bar{2}0 \rangle$	6

SLIP IN SINGLE CRYSTALS:

- A metal single crystal has a number of different slip systems that are capable of operating.
- The resolved shear stress normally differs for each one because the orientation of each relative to the stress axis (ϕ and λ angles) also differs. However, one slip system is generally oriented most favorably—that is, has the largest resolved shear stress, $\tau R(\max)$:
- $\tau R(\max) = \sigma(\cos \phi \cos \lambda)_{\max}$
- In response to an applied tensile or compressive stress, slip in a single crystal commences on the most favorably oriented slip system when the resolved shear stress reaches some critical value, termed the **critical resolved shear stress** τ_{crss} ;
- it represents the minimum shear stress required to initiate slip and is a property of the material that determines when yielding occurs. The single crystal plastically deforms or yields when $\tau R(\max) = \tau_{crss}$, and the magnitude of the applied stress required to initiate yielding (i.e., the yield strength σ_y) is
- $\sigma_y = \tau_{crss} / (\cos \phi \cos \lambda)_{\max}$
- The minimum stress necessary to introduce yielding occurs when a single crystal is oriented such that $\phi = \lambda = 45^\circ$; under these conditions,
- $\sigma_y = 2\tau_{crss}$



- For a single-crystal specimen that is stressed in tension, deformation is as in Figure 1, where slip occurs along a number of equivalent and most favorably oriented planes and directions at various positions along the specimen length. This slip deformation forms as small steps on the surface of the single crystal that are parallel to one another and loop around the circumference of the specimen as indicated in Figure 1.
- Each step results from the movement of a large number of dislocations along the same slip plane. On the surface of a polished single-crystal specimen, these steps appear as lines, which are called *slip lines*. A schematic depiction of slip lines on a cylindrical specimen that was plastically deformed in tension is shown in Figure 2.
- With continued extension of a single crystal, both the number of slip lines and the slip step width increase. For FCC and BCC metals, slip may eventually begin along a second slip system, the system that is next most favorably oriented with the tensile axis.
- Furthermore, for HCP crystals having few slip systems, if the stress axis for the most favorable slip system is either perpendicular to the slip direction ($\lambda = 90^\circ$) or parallel to the slip plane ($\phi = 90^\circ$), the critical resolved shear stress is zero. For these extreme orientations, the crystal typically fractures rather than deforms plastically.



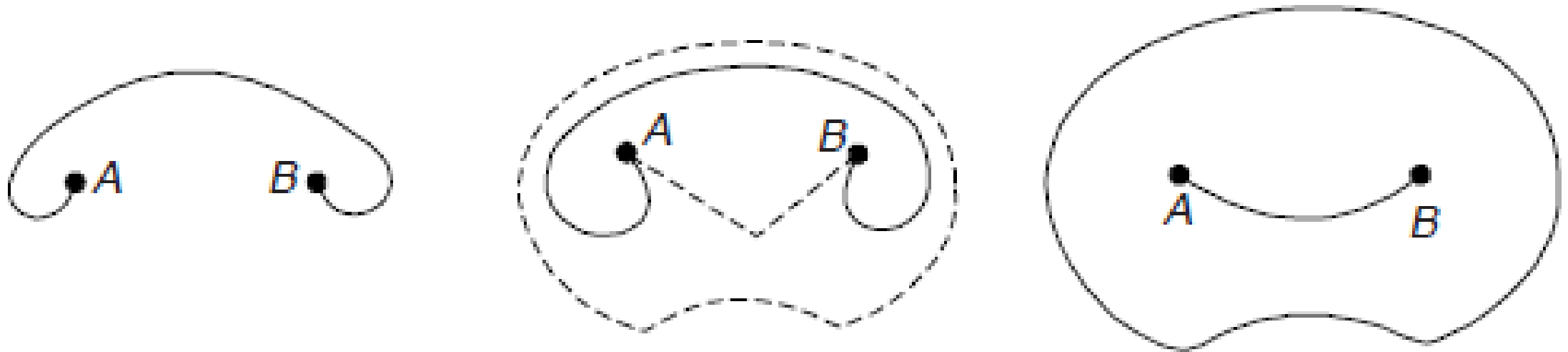
Frank Read Source:

- It is observed that slip occurs intensely on a small number of crystal planes and, during the process, some hundreds of dislocations move.
- Obviously, there must be some effective creators or some mechanisms by which these numerous dislocations are produced on a given slip plane.
- These are called as Frank-Read sources.
- A single Frank-Read dislocation source can form hundreds of new dislocations.
- Frank-Read mechanism helps to explain the existence of so called dislocation mills or multiplication of dislocations

- Let us consider a dislocation line, shown by AB in a crystal. We see that the dislocation line or Frank-Read source consists of two nodes A and B .
- In the first situation, when the Burger vector is perpendicular to the line AB , a shear stress parallel to the plane of the figure will exert a force on the dislocation line AB . Due to the action of the shear stress the dislocation line bent outward and produces slip.
- A slip plane usually contains tens of dislocations. For a given stress, the dislocation line AB assume a certain radius of curvature. On further increasing the stress, the dislocation line becomes unstable and expands indefinitely. Figure below illustrates the successive stages.
- Under the action of the applied stress, the fixed dislocation line AB is bent outward until it becomes hemispherical (Fig. (a) and (b)) of radius $AB/2$.
- Note that if the stress is removed at any stage upto this point AB will regain its original shape.



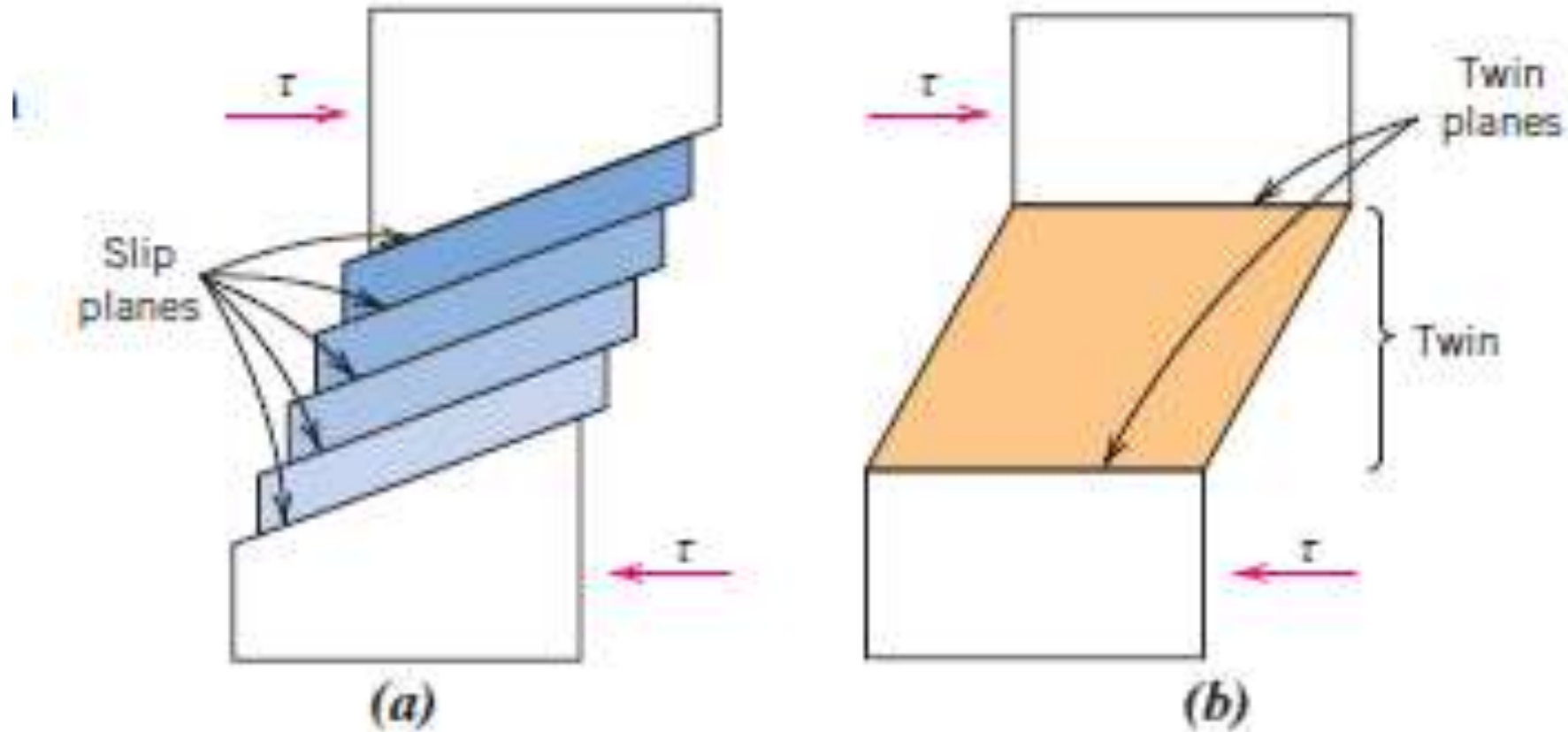
- If the stress is further increased to the stage (Fig. below)) at which the bulge becomes greater than a semicircle, a new system of balance exists and a lower strain energy will be attained by the loop becoming larger (i.e., radius of loop increasing again).
- From that moment on, the bent dislocation propagates spontaneously in the form of two spirals.
- As the spirals meet, they give rise to an expanding dislocation loop and a dislocation section. The dislocation section occupies an initial position and the dislocation source is ready to repeat the cycle.
- A single Frank-Read dislocation source can form hundreds of new dislocations.
- The strengthening effect produced in metals by deformation (strain hardening) is based primarily on an increase of dislocation density.



DEFORMATION BY TWINNING:

- In addition to slip, plastic deformation in some metallic materials can occur by the formation of mechanical twins, or *twinning*.
- The concept of a twin is a shear force can produce atomic displacements such that on one side of a plane (the twin boundary), atoms are located in mirror-image positions of atoms on the other side. The manner in which this is accomplished is demonstrated in Figure .
- Blue circles in Figure *b* represent atoms that did not move—red circles those that were displaced during twinning; magnitude of displacement is represented by red arrows. Furthermore, twinning occurs on a definite crystallographic plane and in a specific direction that depend on crystal structure. For example, for BCC metals, the twin plane and direction are (112) and [111], respectively.
- Slip and twinning deformations are compared in Figure for a single crystal that is subjected to a shear stress τ . Slip ledges are shown in Figure *a*; . For twinning, the shear deformation is homogeneous (Figure *b*). These two processes differ from each other in several respects. First, for slip, the crystallographic orientation above and below the slip plane is the same both before and after the deformation; for twinning, there is a reorientation across the twin plane. In addition, slip occurs in distinct atomic spacing multiples, whereas the atomic displacement for twinning is less than the interatomic separation.

For a single crystal subjected to a shear stress τ , (a) deformation by slip; (b) deformation by twinning.

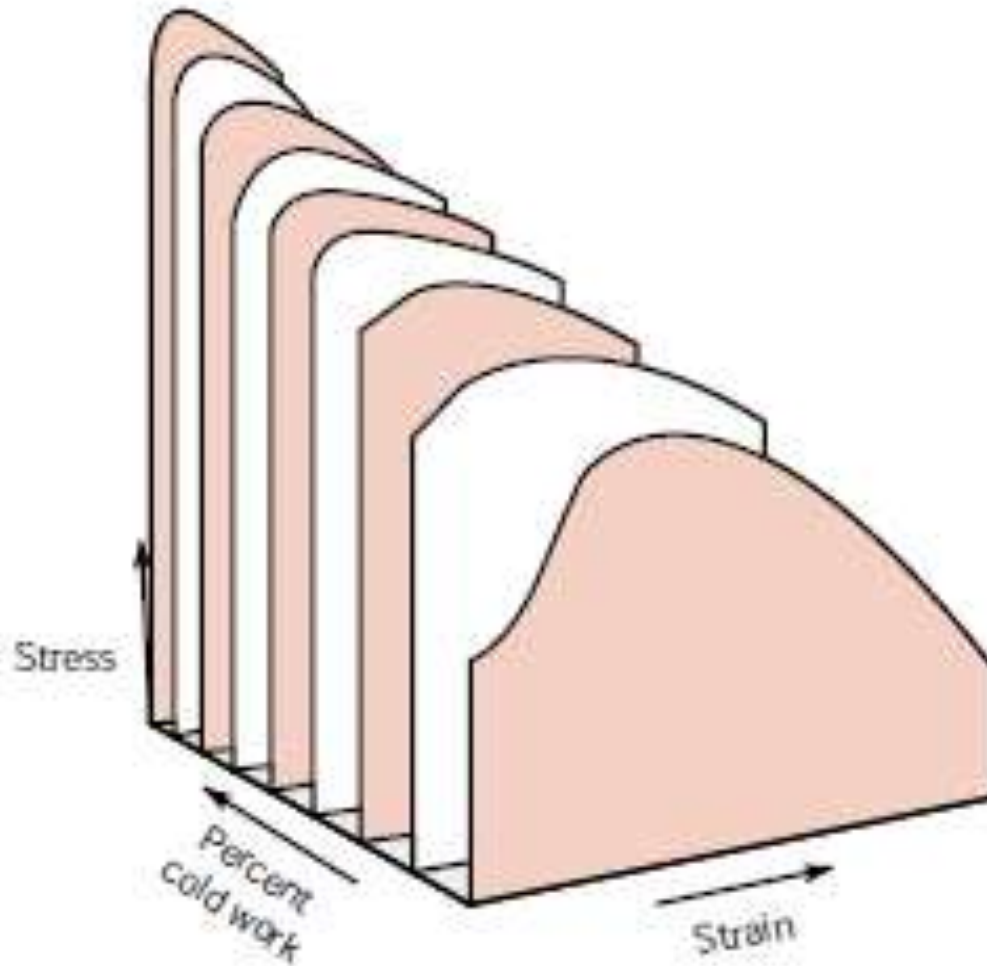


STRAIN HARDENING

- **Strain hardening** is the phenomenon whereby a ductile metal becomes harder and stronger as it is plastically deformed.
- Sometimes it is also called *work hardening*, or because the temperature at which deformation takes place is “cold” relative to the absolute melting temperature of the metal, **cold working**.
- Most metals strain harden at room temperature.

Strain hardening

- The strain hardening phenomenon is explained on the basis of dislocation–dislocation strain field interactions. The dislocation density in a metal increases with deformation or cold work, due to dislocation multiplication or the formation of new dislocations, takes place .
- Consequently, the average distance of separation between dislocations decreases—the dislocations are positioned closer together. On the average, dislocation–dislocation strain interactions are repulsive. The net result is that the motion of a dislocation is hindered by the presence of other dislocations.
- As the dislocation density increases, this resistance to dislocation motion by other dislocations becomes more pronounced. Thus, the imposed stress necessary to deform a metal increases with increasing cold work.
- Strain hardening is often utilized commercially to enhance the mechanical properties of metals during fabrication procedures. The effects of strain hardening may be removed by an annealing heat treatment.



The influence of cold work on
stress strain behaviour of
carbon steel

Effect on mechanical properties :

1. Stress-strain curve
2. Strength and Toughness
3. Hardness
4. Impact
5. Ductility
6. Brittleness.
7. Fatigue
8. Creep
9. Fracture

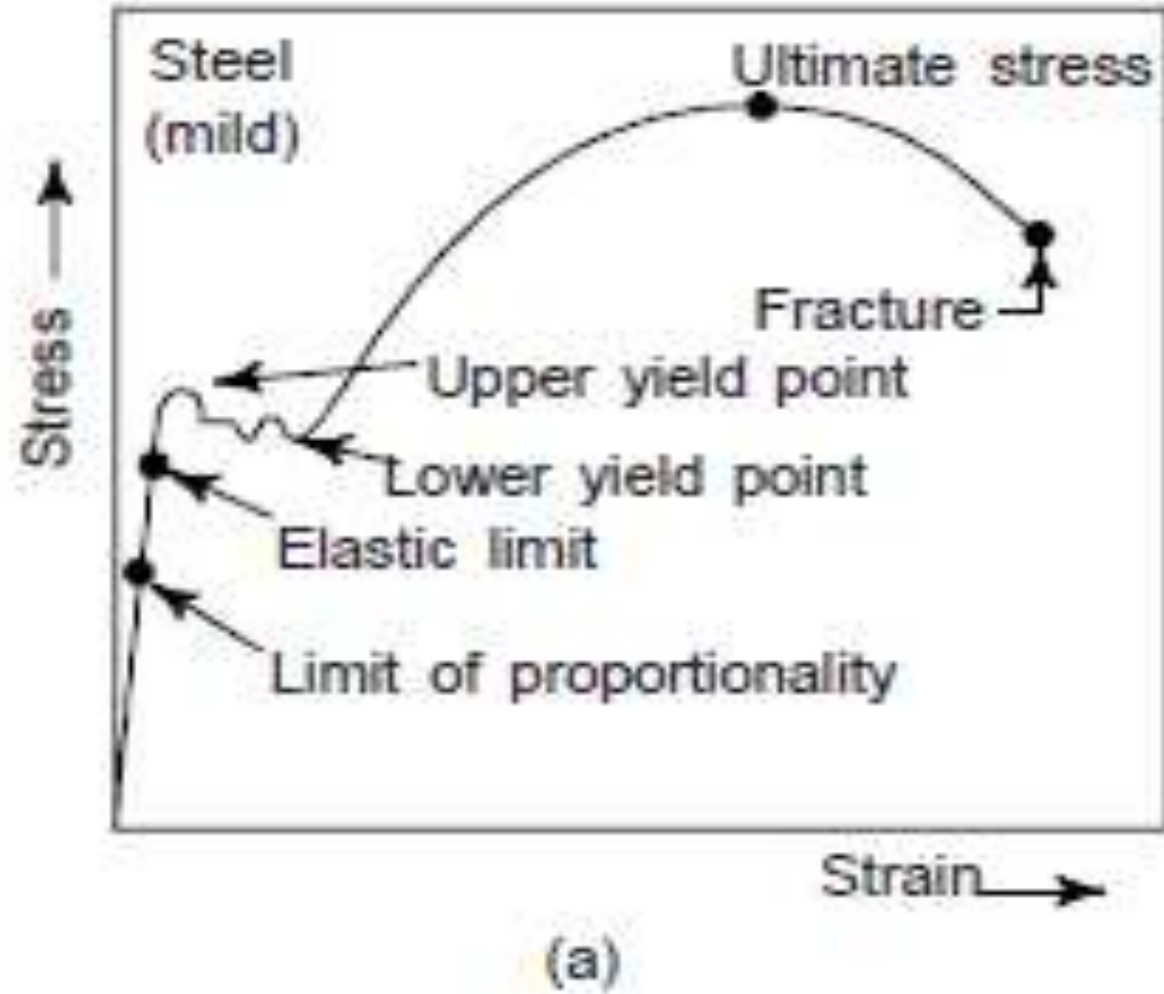
Introduction:

- In determining the fabrication and possible practical applications, the mechanical properties of materials, their strength, rigidity and ductility are of vital importance.
- The important mechanical properties of materials are: *elasticity, plasticity, strength, ductility, hardness, brittleness, toughness, stiffness, resilience, malleability, fatigue, creep*, etc.
- The complete specifications of mechanical properties and composition of various materials have been standardized by BIS.
- Materials exhibit a wide range of mechanical properties ranging, e.g., from the ductility of pure copper to the hardness of diamond and most surprising elastic behaviour of rubber.
- There are several materials behave quite differently when stressed in different ways, e.g. steel and wood are stronger in tension whereas cast iron, cement and bricks are much stronger in compression. Obviously, stresses can produce a shape change and may also cause a material to break or fracture. For materials which have to withstand high temperatures, the melting point is also of importance.

Stress-Strain Diagram

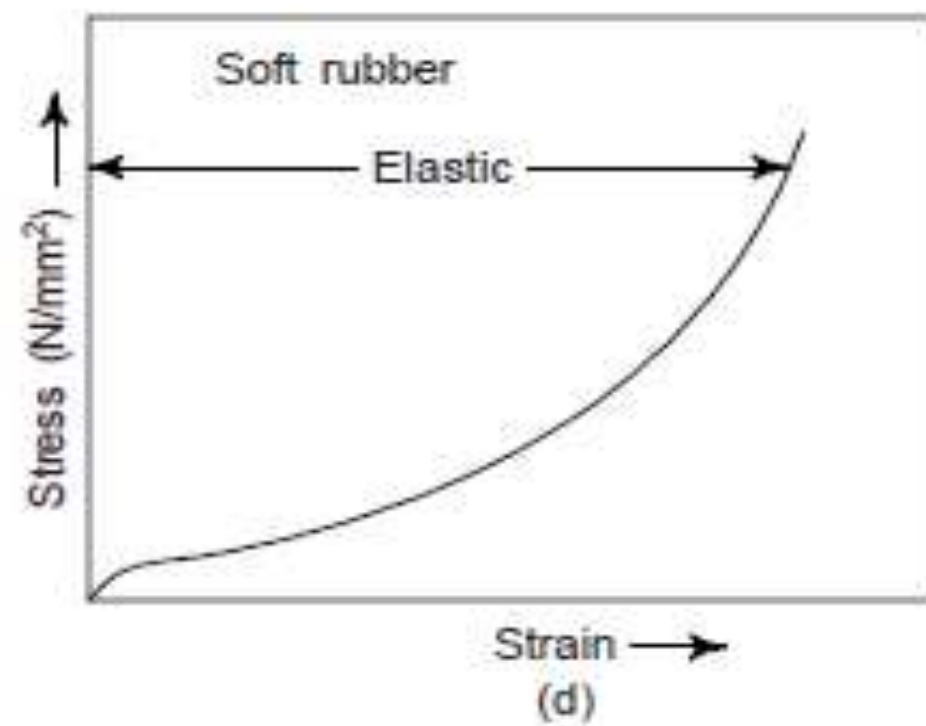
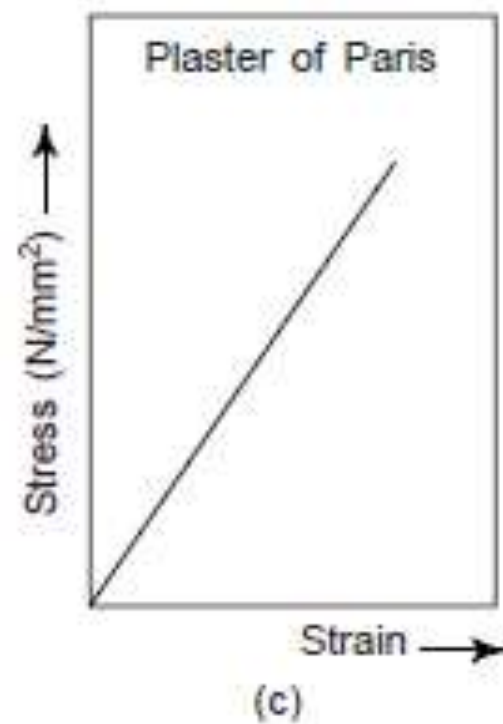
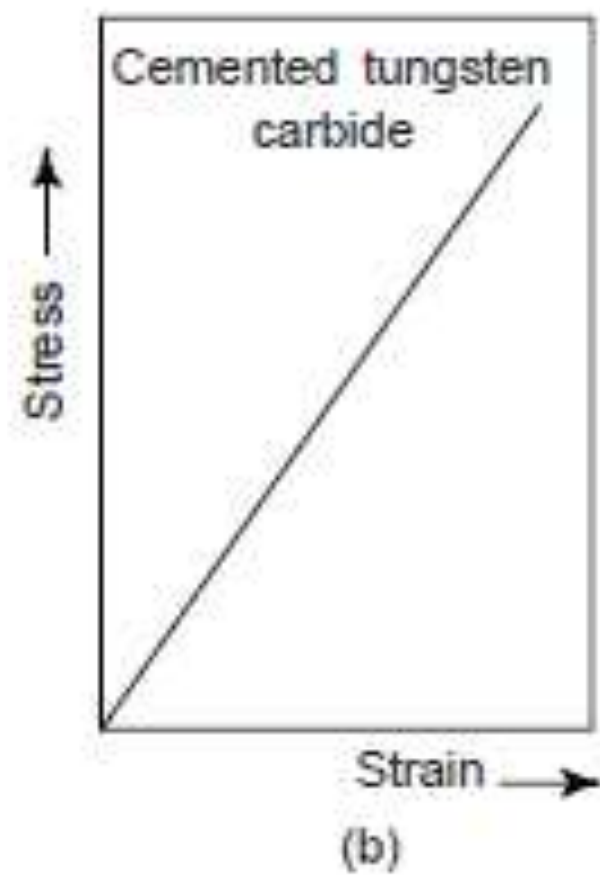
- The stress and strain relation is commonly shown by means of a stress-strain diagram.
- These diagrams are obtained by drawing a graph or curve from the data obtained in a tensile test, in which an increasing tensile stress is applied to a specimen .
- There are resulting changes in length which can be observed and recorded by strain measuring devices. Stress-strain for different engineering materials are shown in Fig.

Stress Strain Curve : Mild Steel



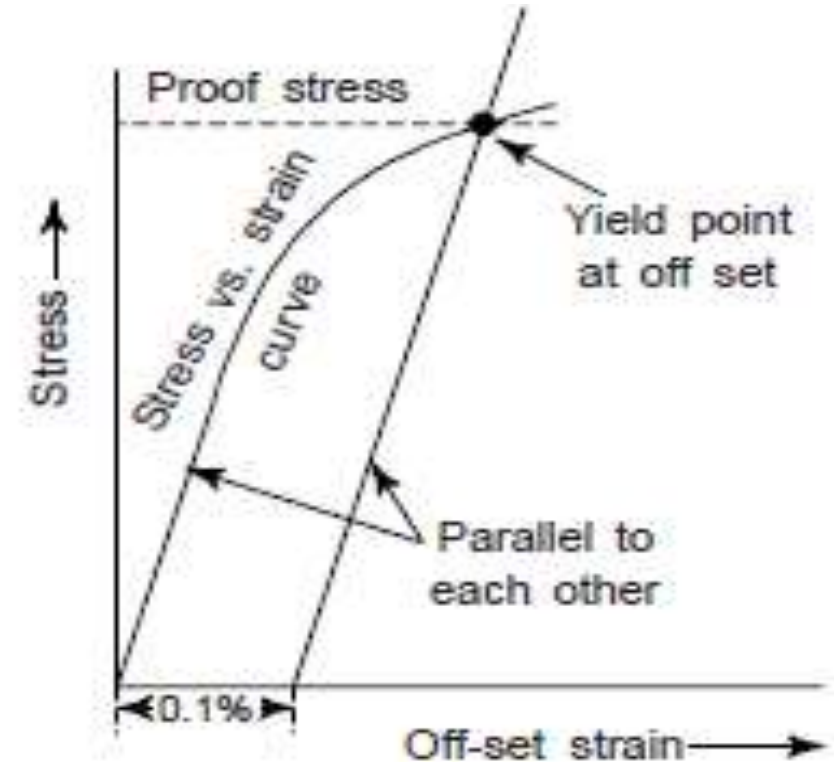
Stress Strain Curve for Mild Materials

- In the case of ductile materials, at the beginning of the test, the material extends elastically. The strain (both longitudinal and lateral) at first increases proportionally to the stress and the sample or specimen returns to its original length on removal of the stress. The limit of proportionality (stress strain) is the stage upto which the specimen, i.e., material obeys Hooke's law perfectly . On further increasing the applied stress, i.e., beyond the elastic limit, it produces plastic deformation so that a permanent extension remains even after the removal of the applied load, i.e. stress. The resultant strain, in this stage begins to increase more quickly than the corresponding stress and continues to increase till the *yield point* is reached.
- We must note that at the yield point the material suddenly stretches. The ratio of applied load to original cross-sectional area is called the *normal stress* and this continues to increase with elongation, due to work hardening or strain hardening, until the tensile stress is maximum.
- This is the value of stress at maximum load and one can calculate it by dividing the maximum load by the original cross-sectional area. This stress is called *ultimate tensile stress* .
- From Fig. it is evident that at a certain value of load the strain continues at slow rate without any further stress or loading. This phenomenon of slow extension increasing with time, at constant stress, is termed *creep*. A neck begins to develop at this point, along the length of the specimen and further plastic deformation is localized within the neck. The cross-sectional area decreases in proportion to the increasing length during elastic elongation. We must note that the volume of the test bar, i.e. specimen remains constant.
- Figure is a stress-strain diagram for mild steel. This diagram clearly shows the limit of proportionality, elastic limit, yield point, ultimate tensile stress and fracture stress at the breaking points. We note that this diagram shows a well-defined yield point.



Stress Strain Curve in Brittle Materials

- Poorly defined yield point as in the case of brittle materials is shown in Fig. For the determination of the yield strength in such materials, following the general practice, one has to draw a straight line parallel to the elastic portion of the stress-strain curve at a predetermined strain ordinate value (say 0.1%).
- The point at which this line intersects the stress vs. strain curve is the yield point at off-set and called the yield strength at 0.1% or 0.2% of set strain. In case of hard steels and non-ferrous metals stress is specified corresponding to a definite amount of permanent elongation. This stress is termed as the proof stress.

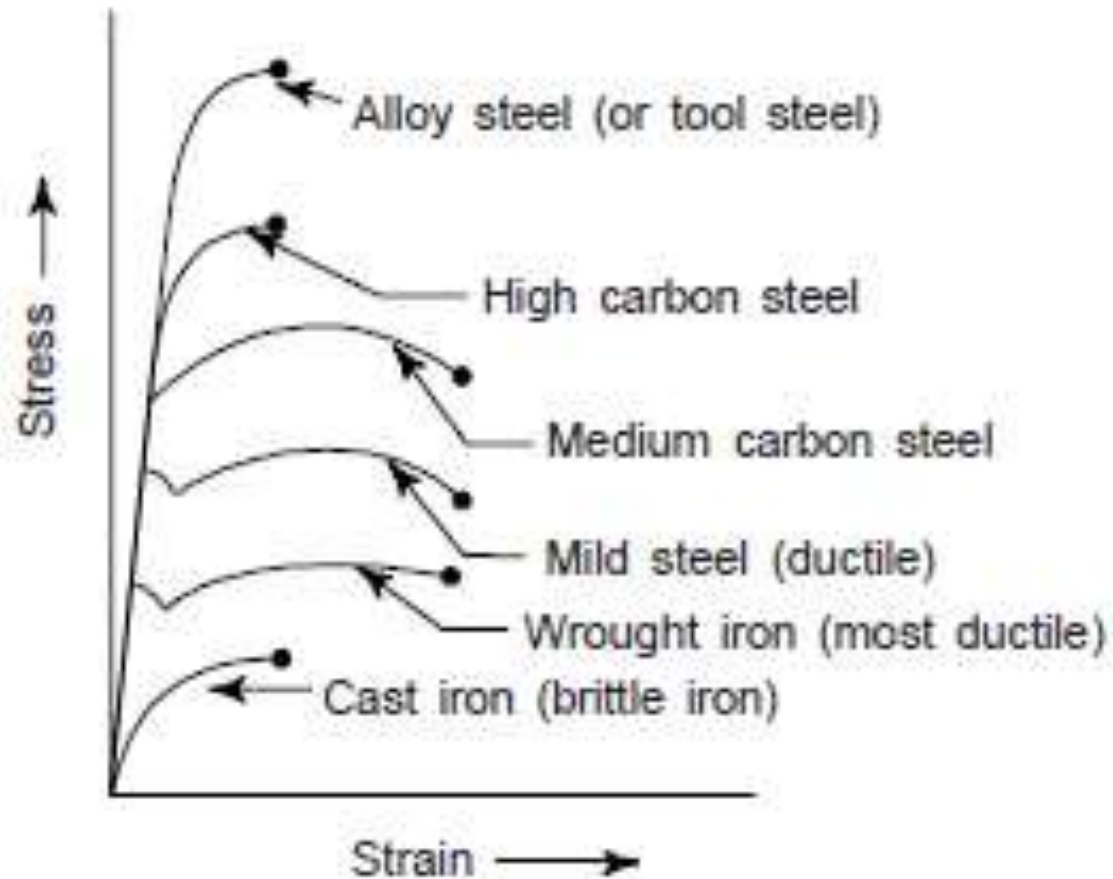


- the proof stress is applied for 15 seconds and, when removed, the specimen should not lengthen permanently beyond 0.1%. The method of finding the proof stress from the stress-strain curve is shown in Fig.

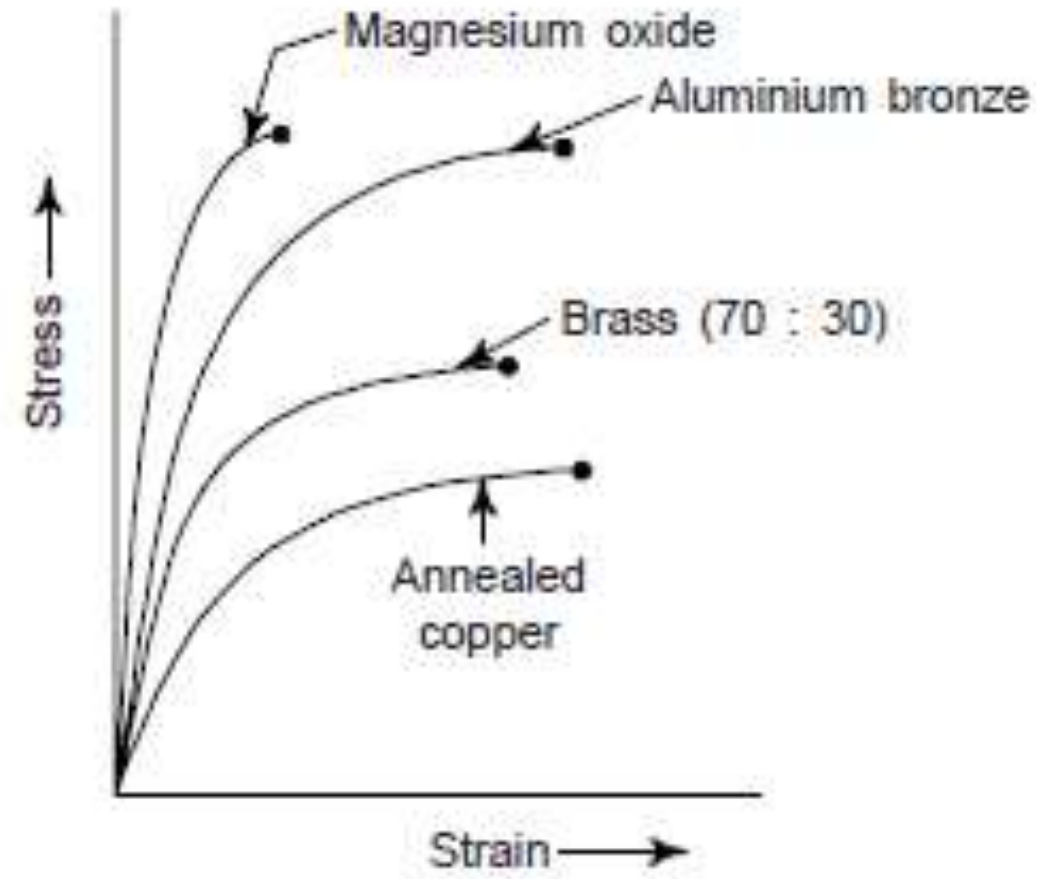
Some Properties Deduce from Stress Strain Curve

- Stress vs. strain curves also help to explain the properties of ductile materials.
- The greater the angle of inclination of the line of stress vs. strain proportionality to the ordinates, the more elastic is that metal.
- A higher yield point reveals greater hardness of the metal.
- A higher value of the maximum stress point shows that the metal is a stronger one.
- Similarly, the toughness and brittleness of metal are indicated by the distance from the ordinates of the breaking stress or load point.
- The metal is more brittle when the distance is shorter.
- Stress vs. strain curves for ferrous and non-ferrous materials are shown in Fig. (Next Slide) .
- The brittle materials show little or no permanent deformation prior to fracture. Some metals and magnesium oxide exhibit brittle behaviour

Stress-Strain Curve for Ferrous & Non Ferrous Materials



(a)



(b)

Strength and Toughness

- It may be defined as the property of a metal by virtue of which it can absorb maximum energy before fracture takes place. It is the measurement of ultimate energy strength of material and is expressed as work units/unit volume, i.e. kg fm/m^3 . Toughness is also calculated in terms of area under stress-strain curve (Fig. (A)).
- We may note that the toughness is maximum under Fig. (a) (c). We may note that the total area under stress-strain curve is the work expended in deforming one m^3 of material until it fractures.
- This work is also termed as *modulus of toughness*. Tenacity and hardness of a metal are the measures of toughness.
- The toughness is approximately measured as the product of strength and elongation and these measurements are used for comparison purposes and termed as *merit number*.

Characteristics of Toughness

- Toughness is the property of materials which enables a material to be twisted, bent or stretched under a high stress before rupture.
- The value of toughness falls with the rise in temperature.
- Toughness is highly desirable property for structural and mechanical parts which have to withstand shock and vibration.
- Its importance lies in the selection of a material where the load increases beyond the elastic limit or yield point, e.g. power press punch and pneumatic hammer etc.
- Amongst the tough materials, few are: brass, mild steel, manganese and wrought.
- A ductile material with same strength as non-ductile material requires more energy for breaking and hence is more tough.

Hardness

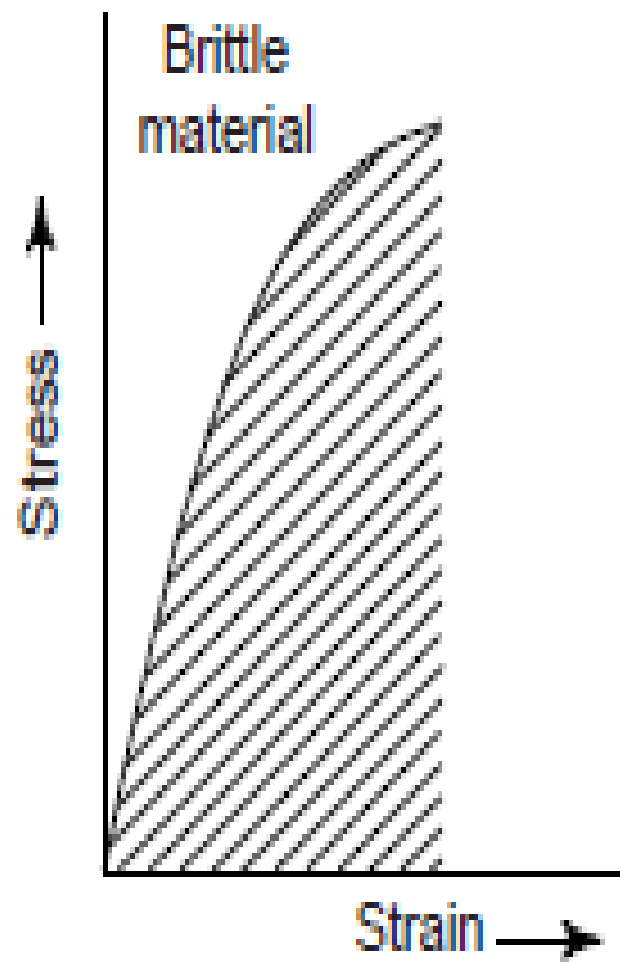
- This is the property of a material (metal) by virtue of it is able to resist abrasion, indentation (or penetration) and scratching by harder bodies. It is the resistance of a material to permanent deformation of the surface.
- In other words, one can define it as the resistance of the metal to penetration by an indenter. The hardness of a surface of the material is, of course, a direct result of interatomic forces acting on the surface of the material. We must note that hardness is not a fundamental property of a material, however, but rather a combined effect of compressive, elastic and plastic properties relative to the mode of penetration, shape of penetrator, etc. Hardness seems to bear a fairly constant relationship to the tensile strength of a given material and thus it can be used as a practical non-destructive test for an approximate idea of the value of that property and the state of the metal near the surface.
- There are several methods of hardness testing, depending either on the direct thrust of some form of penetrator into the metal surface, or on the ploughing of the surface as a stylus is drawn across it under a controlled load, or on the measurement of elastic rebound of an impacting hammer which possessing known energy. Measurements of hardness are the easiest to make and are widely used for industrial design and in research. All hardness tests are made on the surface or close to it. We may note that in mechanical tests the bulk of material is involved.
- Brinell hardness number is the hardness index, calculated by pressing a hardened steel ball (indenter) into test specimen under standard load. Brinell, Rockwell and Vickers hardness tests are used to determine hardness of metallic materials, to check quality level of products, for uniformity of samples of metals, for uniformity of results of heat treatment.

Brittleness:

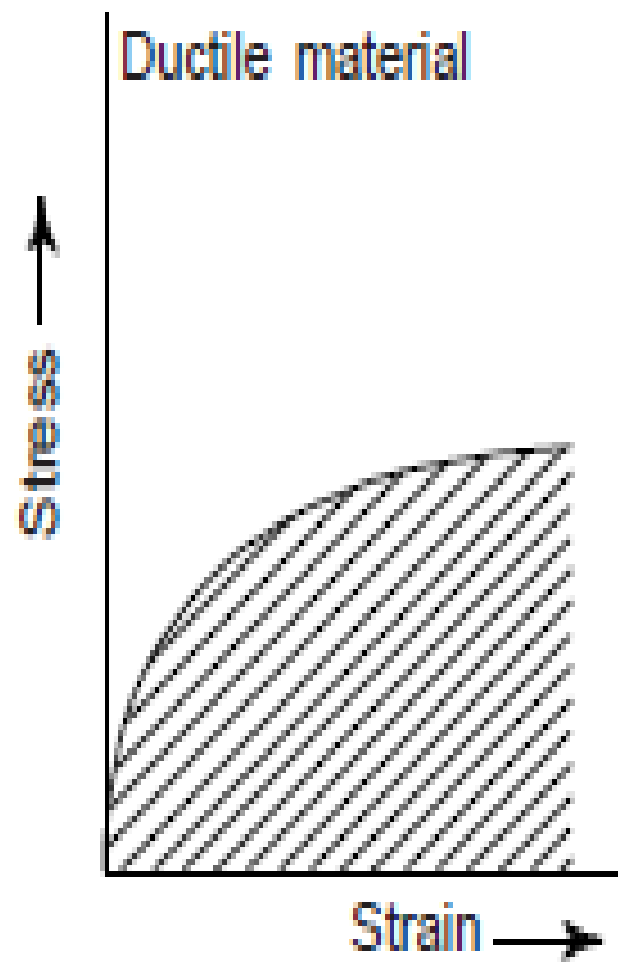
- It may be defined as the property of a metal by virtue of which it will fracture without any appreciable deformation.
- This property is just opposite to the ductility of a metal. Few examples of brittle materials are cast iron, glass and concrete.
- This property of metals find its importance for design of machine tools, which are subjected to sudden loads.
- Metals with less than 5% elongation are known to be brittle ones.

Poisson's ratio (ν)

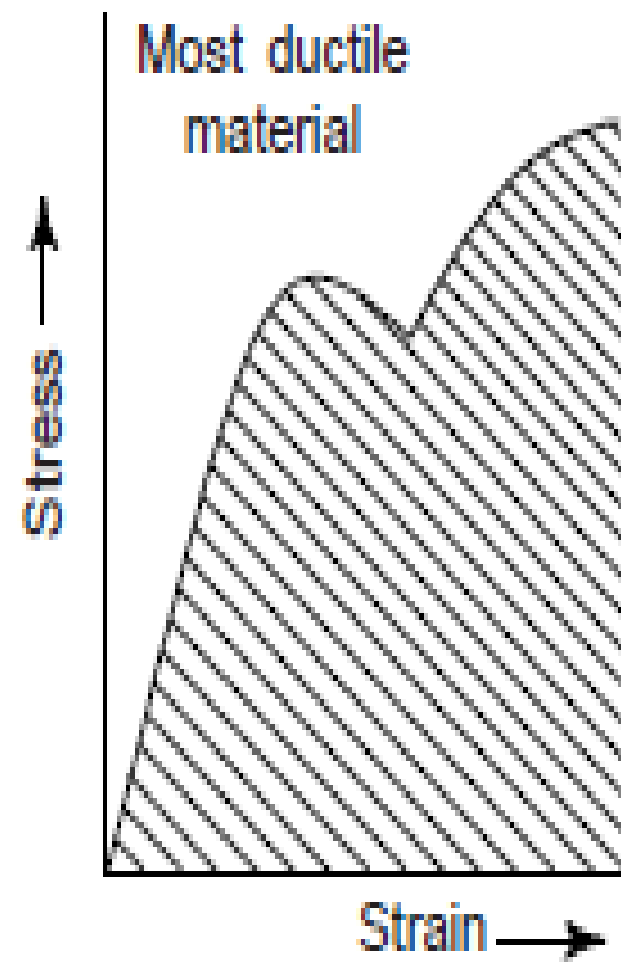
- When a specimen is stressed by a uniaxial force, it is strained in the direction of the force and also in a direction perpendicular to the direction of the force. The strain in the direction of force is called longitudinal strain and that perpendicular to it as lateral strain. The relationship existing between the lateral strain and axial strain is called Poisson's ratio (ν):
- Poisson's ratio (ν) = Lateral strain / Longitudinal strain = l/m = (constant)
- Poisson's ratio is an important elastic constant and its value is different for different materials. For most engineering structural materials its value is usually between 0.3 and 0.5. The relation between E , G and ν is
- $\nu = \frac{E}{2G} - 1$



(a)



(b)



(c)

Stiffness:

- This may be defined as the property of a metal by virtue of which it resists deformation.
- Modulus of rigidity is the measure of stiffness. The term flexibility is quite opposite of stiffness. The materials which suffer less deformation under load have high degree of stiffness. We may note that the greater the stress required to produce a given strain, the stiffer is the material.
- The stiffness of a metal is of great importance while selecting it for a number or a component of a machine or a structure. To make it clear, let us consider an example of two beams one of steel and other of aluminium. Both the beams are strong enough to carry the given load but aluminium beam will sag or deflect more than the mild steel beam. Obviously, the steel beam is stiffer than the aluminium beam.
- Stiffness is also used in graduating spring balance and spring controlled measuring instruments. Stiffness or flexibility is the function of dimensions, shape and characteristics of the material.

Resilience:

- This may be defined as the property of a metal by virtue of which it stores energy and resists shocks or impacts. It is measured by the amount of energy absorbed per unit volume, in stressing a material upto elastic limit.
- This property is of great importance in the selection of a material used for various types of springs.
- The maximum energy which can be stored in a body upto elastic limit is termed as *proof resilience*.
- Proof resilience per unit volume is termed as *modulus of resilience*. Obviously, the energy stored per unit volume at elastic limit is modulus of resilience.
- The annealed copper is not used for springs due to its very low elastic limit. However, the cold-worked copper has much higher elastic limit (and resilience) and therefore it is used for springs. This shows that resilience is associated with high elastic limit. We may note that resilience is of importance for materials required to bear shocks and vibrations.

Endurance:

- This is defined as the property of a metal by virtue of which it can withstand varying stresses (same or opposite nature).
- The maximum value of stress, which can be applied for an indefinite times without causing its failure, is termed as its *endurance* limit.
- For ordinarily steel, the endurance limit is about half the tensile strength.
- This property of a metal is of great importance in the design and production of parts in a reciprocating machines and components subjected to vibrations.
- It is always desirable to keep the working stress of a material well within the elastic limit.
- It is Most Important Property in Designing Movable parts

Heat Treatment of Steels

Contents

- Phase transformation.
- TTT diagrams (S curves).
- Stress relief and Annealing.
- Quenching and Hardening.
- Tempering.
- Normalizing.
- Hardenability.
- Case hardening.
- Carburising.
- Nitriding.
- Cyaniding.

Heat Treatment



Heat Treatment

- The amount of carbon present in plain carbon steel has a pronounced effect on the properties of a steel and on the selection of suitable heat treatments to attain certain desired properties.
- Heat treatment may be defined as:
- An operation or combination of operations involving Heating and cooling of a metal//alloy in solid state to obtain desirable

Conditions, e.g., that of
relieved stresses

Properties, e.g.,
- Better machinability
- Improved ductility
- Homogeneous structure, etc.

Purpose

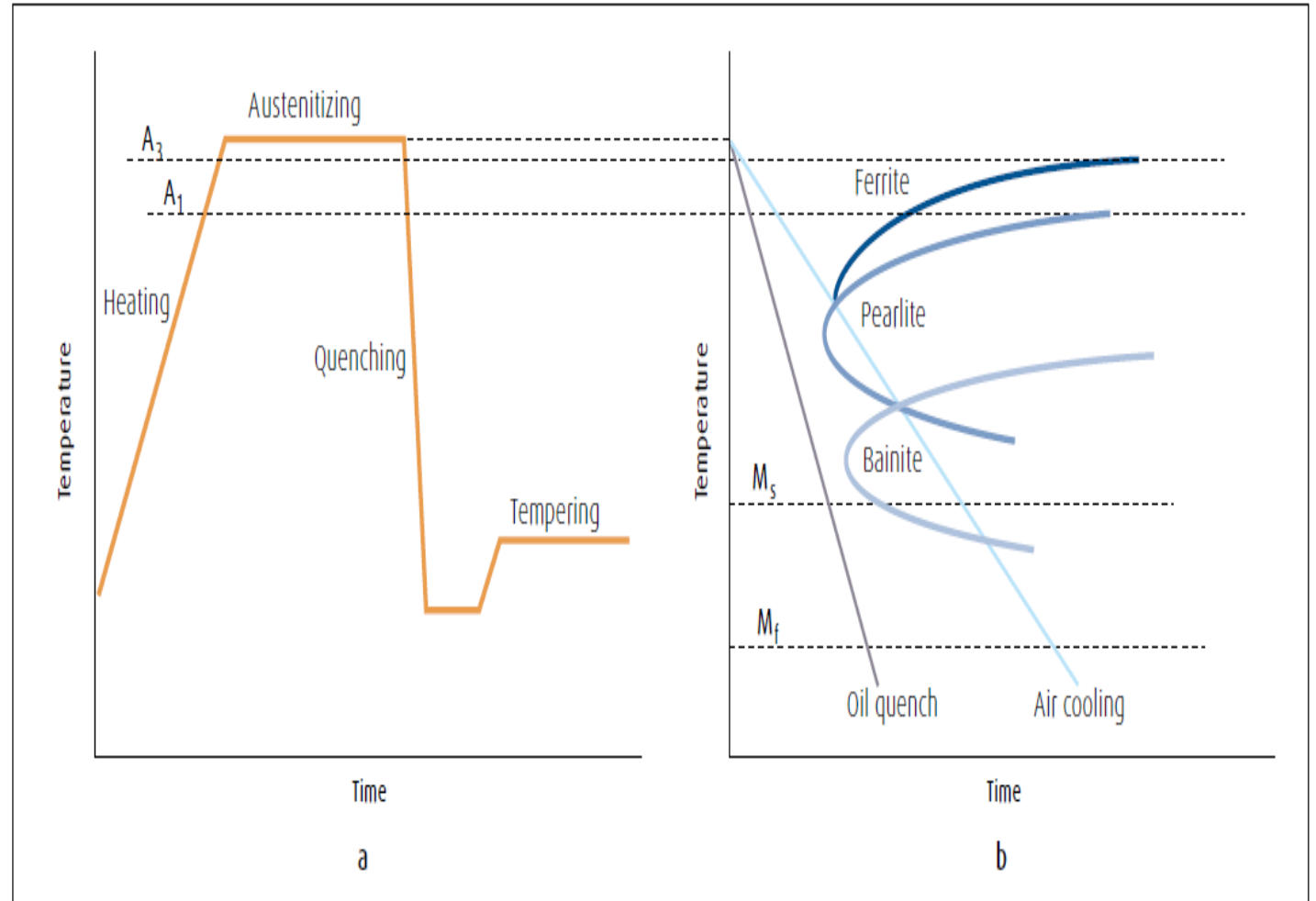
- Cause relief of internal stresses developed during cold working, welding, casting, forging etc.
- Harden and strengthen metals.
- Improve machinability.
- Change grain size
- Soften metals for further (cold) working as in wire drawing or cold rolling.
- Improve ductility and toughness
- Increase , heat, wear and corrosion resistance of materials.
- Improve electrical and magnetic properties.
- Homogenize the structure to remove coring .
- Spheroidize tiny particles, such as those of Fe_3C in steel, by diffusion.

Heat Treatment Theory

- The various types of heat-treating processes are similar because they all involve the heating and cooling of metals.
- they differ in the heating temperatures and the cooling rates used and the final results.
- Ferrous metals (metals with iron) are annealing, normalizing, hardening, and tempering.
- Nonferrous metals can be annealed, but never tempered, normalized, or case-hardened.

Stages of Heat Treatment

- **Stage 1**—Heating the metal slowly to ensure a uniform temperature.
- **Stage 2**—Soaking (holding) the metal at a given temperature for a given period of time.
- **Stage 3**—Cooling the metal to room temperature.



a) Temperature-time cycle and b) its relation to the TTT diagram

- **Soaking:** - Internal structural changes take place. soaking period depends on the chemical analysis of the metal and the mass of the part.
- **Cooling Stage:** - To cool the metal, you can place it in direct contact with a COOLING MEDIUM composed of a gas, liquid, solid, or combination of these.
- **Soaking Period:** Table 1: Soaking period for Hardening, Annealing and Normalizing Steel.

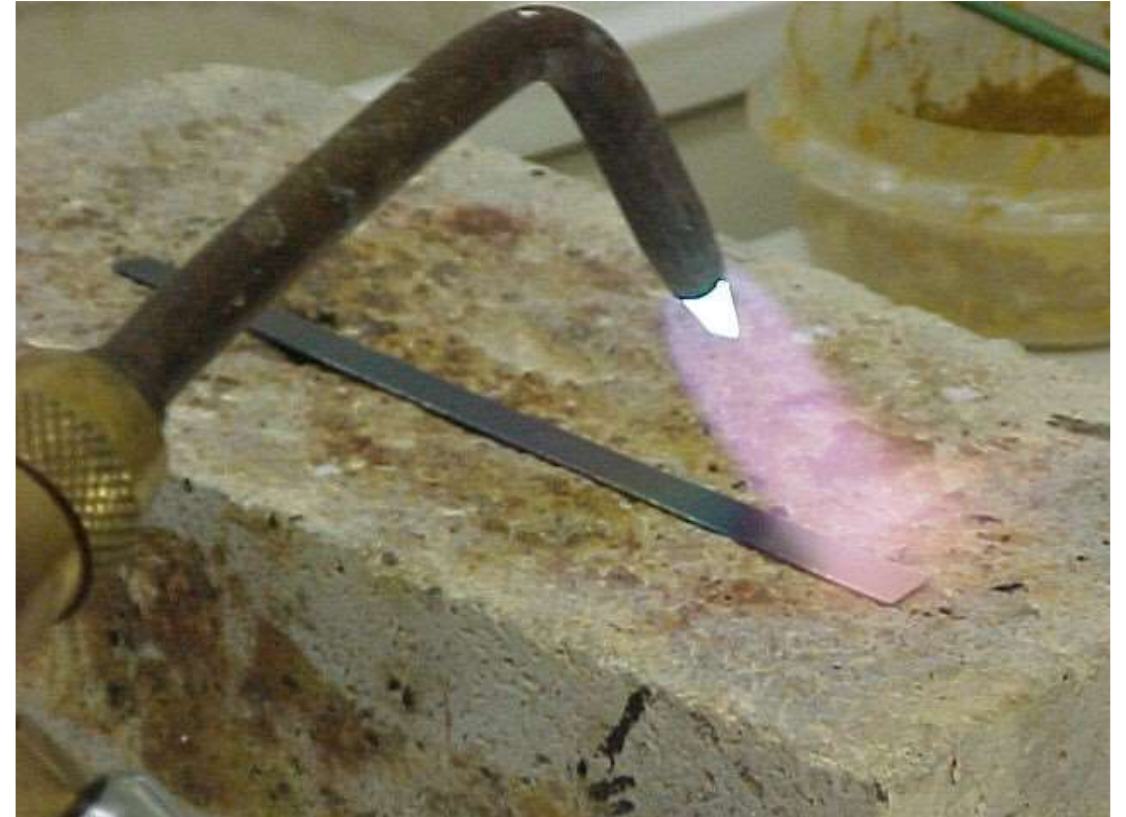
Thickness of Metal (inches)	Time of heating to Required Temperature (hr)	Soaking Time (hr)
Up to 1	3/4	1/2
1 to 2	1 1/4	1/2
2 to 3	1 3/4	3/4
3 to 4	2 1/4	1
4 to 5	2 3/4	1
5 to 8	3 1/2	1 1/2

Types

- *Annealing*
- *Normalizing*
- *Hardening*
- Carburizing
- *Tempering*

Annealing

- Steel is annealed to reduce the hardness, improve machine ability, facilitate cold-working, produce a desired microstructure.
- Full annealing is the process of softening steel by a heating and cooling cycle, so that it may be bent or cut easily.
- In annealing, steel is heated above the transformation temperature to form austenite, and cooled very slowly, usually in the furnace.



CLASSIFICATION

- Annealing treatment can be classified into groups based on the following:

1. Heat treatment temperature	2. Phase transformation	3. Specific purpose
• Full annealing • Partial annealing • Sub-critical annealing	• First-order annealing • Second-order annealing	• Full annealing • Isothermal annealing • Diffusion annealing • Partial annealing • Recrystallization annealing • Spheroidisation annealing

Annealing Based on Temperature

- In full annealing the steel is heated above the critical temperature(A_3) and then cooled very slowly.
- Partial annealing, also known as incomplete annealing or intercritical annealing, involves heating of steel to a temp. lying between lower critical temperature(A_1) and upper critical temperature (A_3 or A_{cm}).
- Subcritical annealing is a process in which the maximum temp. to which is heated is always less than the lower critical temperature(A_1).

Classification based on phase transformation features.

- **First-order annealing** is performed on steel with the sole aim of achieving some properties.
- Any change in the characteristics of steel achieved by this type of annealing is not correlated to phase transformation.
- it can be performed at a wide range of temperatures above or below the critical temperatures.
- The **second-order annealing** differs from the former in the sense that the end results in the former are essentially due to phase transformation which takes place during the treatment.

Types of annealing based on specific purpose

3.1 Full annealing

- In this, steel is heated to its 50°C above the austenitic temperature and held for sufficient time to allow the material to fully form austenite or austenite-cementite grain structure. The material is then allowed to cool slowly so that the equilibrium microstructure is obtained.
- The austenitising temp is a function of carbon content of the steel and can be generalized as:
 - For hypoeutectoid steels and eutectoid steel
 - $\text{Ac}_3 + (20-40^{\circ}\text{C})$ [to obtain single phase austenite]
 - For hypereutectoid steels
 - $\text{Ac}_1 + (20-40^{\circ}\text{C})$ [to obtain austenite+ cementite]

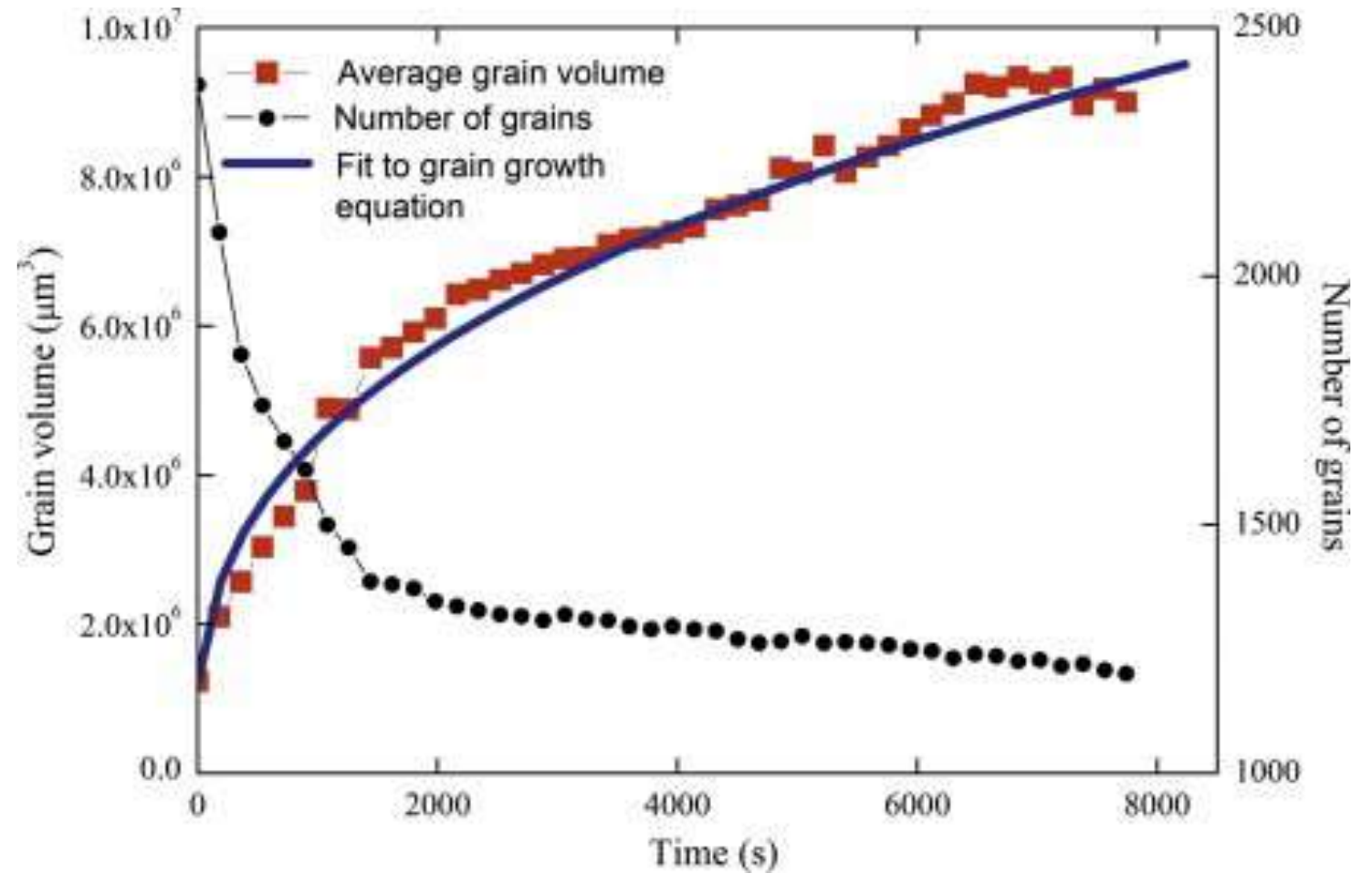
Purpose of full annealing

- To relieve internal stresses
- To reduce hardness and increase ductility
- For refining of grain size
- For making the material having homogeneous chemical composition
- For making the material suitable for high machining processes
- To make steel suitable for undergoing other heat treatment processes like hardening, normalizing etc.

Isothermal Annealing

- It is a process in which hypoeutectoid steel is heated above the upper critical temperature and this temperature is maintained for a time and then the temperature is brought down below lower critical temperature and is again maintained. Then finally it is cooled at room temperature.
- The prefix “isothermal” associated with annealing implies that transformation of austenite takes place at constant temperature.

The closer the temp of isothermal holding is to A₁, coarser is the pearlite, softer is the steel, but longer is the time of isothermal transformation.



- Advantages:
 - Improved machinability.
 - Homogeneous structure and better surface finish.
 - Time required for complete cycle is comparably less.
 - The process is of great use for alloy steels as the steels have to be cooled slowly.
- Limitation:
 - It is suitable only for small-sized components.
 - Heavy components cannot be subjected to this treatment because it is not possible to cool them rapidly and uniformly to the holding temperature at which transformation occurs. Thus structure won't be homogeneous. so mechanical properties will vary across the cross-section.

Diffusion Annealing

- This process, also known as homogenizing annealing, is employed to remove any structural non-uniformity like dendrites, columnar grains and chemical inhomogeneity which promote brittleness and reduce ductility and toughness of steel.
- Process:
 - Steel is heated sufficiently above the upper critical temperature (say, 1000-2000°C), and held at this temperature for 10-20 hours, followed by slow cooling.
 - Segregated zones are eliminated, and a chemically homogeneous steel is obtained by this treatment as a result of diffusion.
 - Heating to such a high temp. results in considerable coarsening of austenitic grains & heavy scale formation. The coarse austenite thus obtained further transforms to coarse pearlite on cooling, which is not a desirable structure as mechanical properties are impaired.

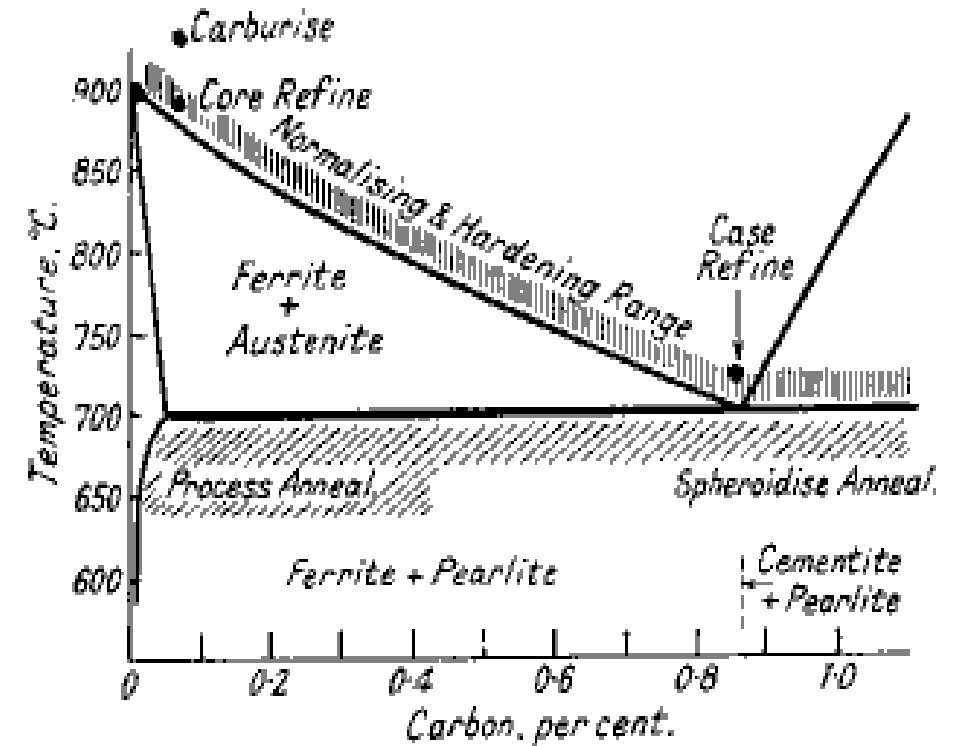
- The main aim of homogenising annealing is to make the composition uniform, i.e to remove chemical heterogeneity.
- The impact energy and ductility of the steel increase as the homogenizing temperature increases and the hardness, yield strength and tensile strength decrease with an increase in the homogenizing temperature.
- Homogenizing annealing has a few shortcomings as well. It results in:
 - Grain coarsening of austenite, thereby impairing the properties
 - Thick scales on the surface of steels
 - It is an expensive process

Partial Annealing

- Partial annealing, also known as inter-critical annealing or incomplete annealing, is a process in which steel is heated between A_1 and A_{cm} and is followed by slow cooling.
- Generally, hypereutectoid steels are subjected to this treatment. The resultant microstructure consists of fine pearlite and cementite instead of coarse pearlite and a network of cementite at grain boundaries.
- As low temperatures are involved in this process, it is less expensive than full annealing.
- Hypoeutectoid steels are subjected to this treatment to improve machinability.
- However, steels with coarse structure of ferrite and pearlite are not suitable for this treatment. This is because only a considerable amount of ferrite remains untransformed, and only a part of it along with pearlite transforms to austenite.
- This coarse or acicular untransformed ferrite results in poor mechanical properties.

Recrystallization Annealing

- The process consists of heating steel above the recrystallization temperature, holding at this temperature and cooling thereafter.
- It is used to treat work-hardened parts made out of low-Carbon steels ($< 0.25\%$ Carbon).
- This allows the parts to be soft enough to undergo further cold working without fracturing.



- Recrystallization temp(T_r) is given by:

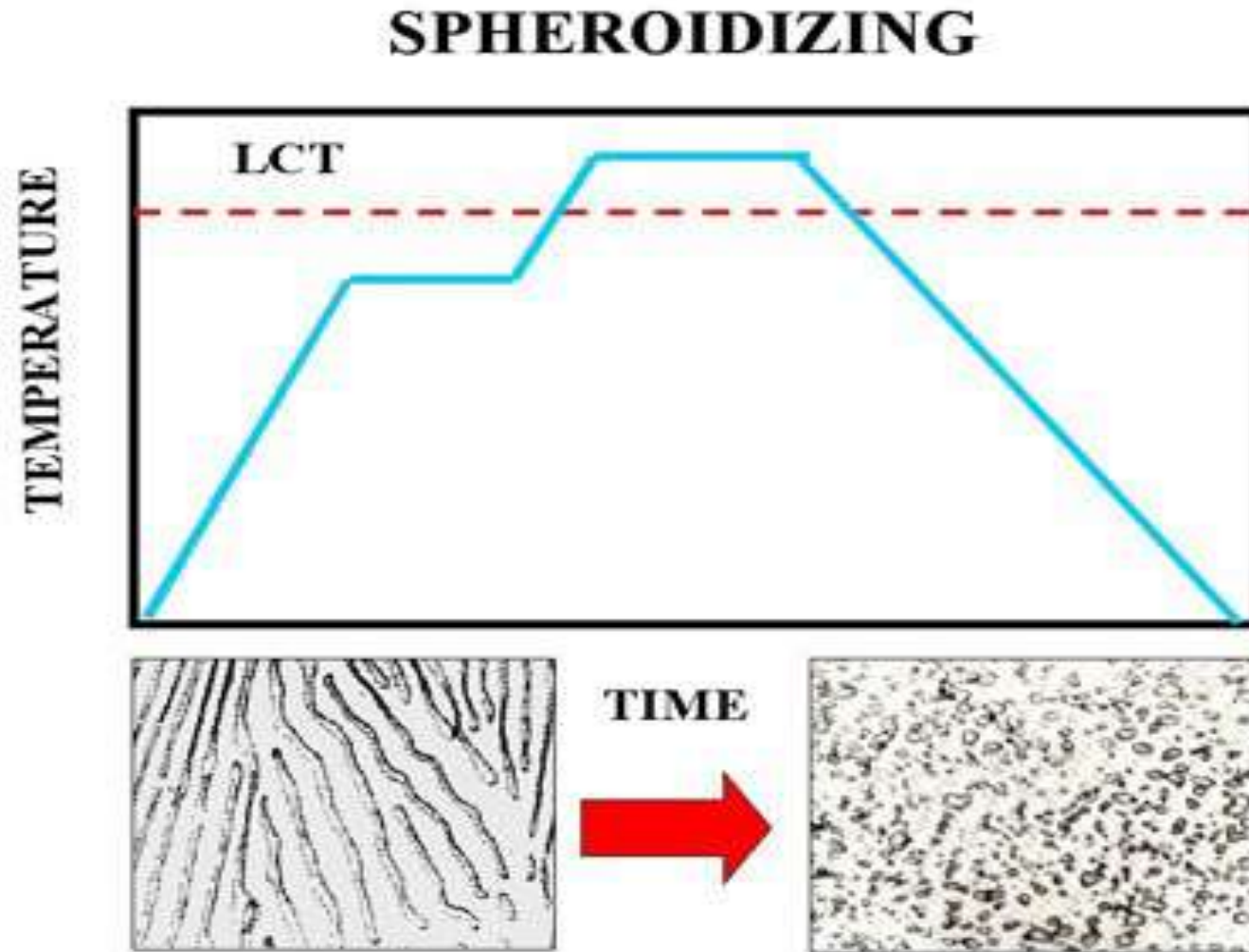
$$T_r = (0.3-0.5)T_{mp} \quad (T_{mp} \text{ is Melting point temperature})$$

- As little scaling and decarburization occurs in recrystallization annealing, it is preferred over full annealing.
- No phase change takes place and the final structure consists of strain-free, equiaxed grains of fine ferrite produced at the expense of deformed elongated ferrite grains.
- However It would produce very coarse grains if the steel has undergone critical amount of deformation. In such cases, full annealing is preferred.
- Aims of Recrystallization Annealing
 - To restore ductility
 - To refine coarse grains
 - To improve electrical and magnetic properties in grain-oriented Si steels.

Spheroidization annealing

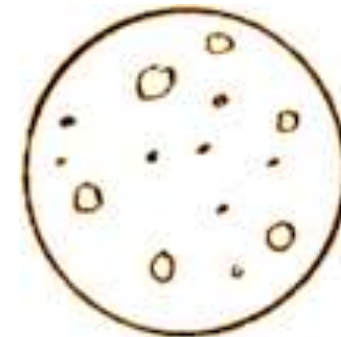
- Spheroidization annealing consists of heating, soaking and cooling, invariably very slowly to produce spheroidal pearlite or globular form of carbides in steels.
- To improve the machinability of the annealed hypereutectoid steel spheroidize annealing is applied.
- Hypereutectoid steels consist of pearlite and cementite. The cementite forms a brittle network around the pearlite. This presents difficulty in machining the hypereutectoid steels.
- This process will produce a spheroidal or globular form of a carbide in a ferritic matrix which makes the machining easy.
- Prolonged time at the elevated temperature will completely break up the pearlitic structure and cementite network. The structure is called spheroidite.

Spheroidizing process applied at a temperature below and above the LCT.



Spheroidising Process:

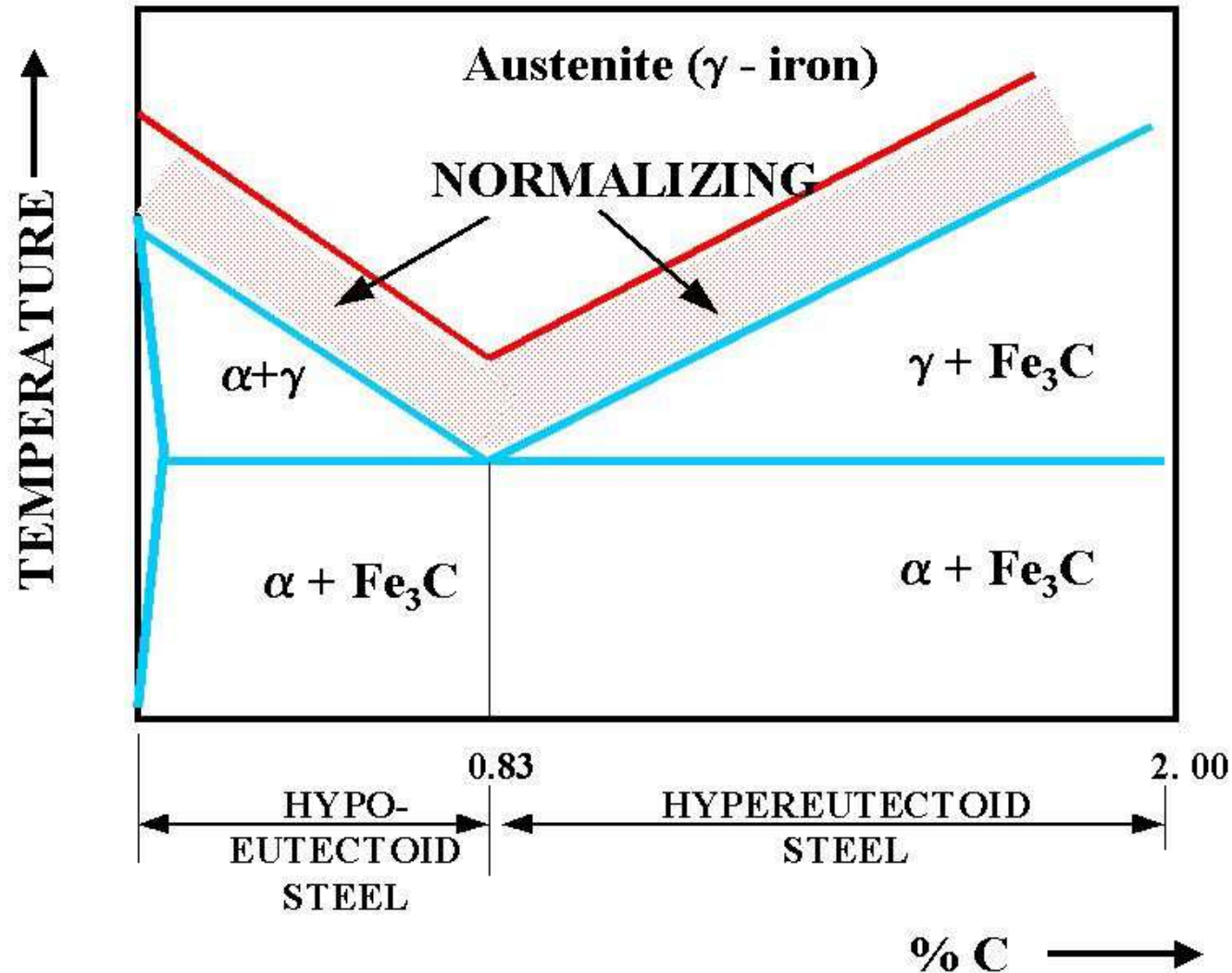
- Heat the part to a temperature just below the Ferrite- Austenite line, line A1 or below the Austenite- Cementite line, essentially below the 727 °C (1340°F) line. Hold the temperature for a prolonged time and follow by fairly slow cooling. Or
- Cycle multiple times between temperatures slightly above and slightly below the 727 °C (1340 °F) line, say for example between 700 and 750 °C (1292 -1382 °F), and slow cool. Or
- For tool and alloy steels heat to 750 to 800 °C (1382- 1472 °F) and hold for several hours followed by slowcooling.
- All these methods result in a structure in which all the Cementite is in the form of small globules (spheroids) dispersed throughout the ferrite matrix.
- This structure allows for improved machining in continuous cutting operations such as lathes and screw machines. Spheroidization also improves resistance to abrasion.
- **Aims of Spheroidization Annealing:**
 - minimum hardness
 - maximum ductility
 - maximum machinability
 - maximum softness



NORMALIZING

- The normalizing of steel is carried out by heating above the UCT (Upper Critical Temperature) to single phase austenitic region to get homogeneous austenite, soaking there for some time and then cooling it in air to room temperature.
- The austenitising temperature range are:
 - For hypoeutectoid steels and eutectoid steel
 $A_{c3} + (40-60\text{ C})$
 - For hypereutectoid steels
 $A_{cm} + (30-50\text{ C})$
- During normalising we use grain refinement which is associated with allotropic transformation upon heating $\gamma \rightarrow \alpha$
- Parts that require maximum toughness and those subjected to impact are often normalized.
- When large cross sections are normalized, they are also tempered to further reduce stress and more closely control mechanical properties.
- The microstructure obtained by normalizing depends on the composition of the castings (which dictates its hardenability) and the cooling rate.

Figure below shows the normalizing temperatures for hypoeutectoid and hypereutectoid steels



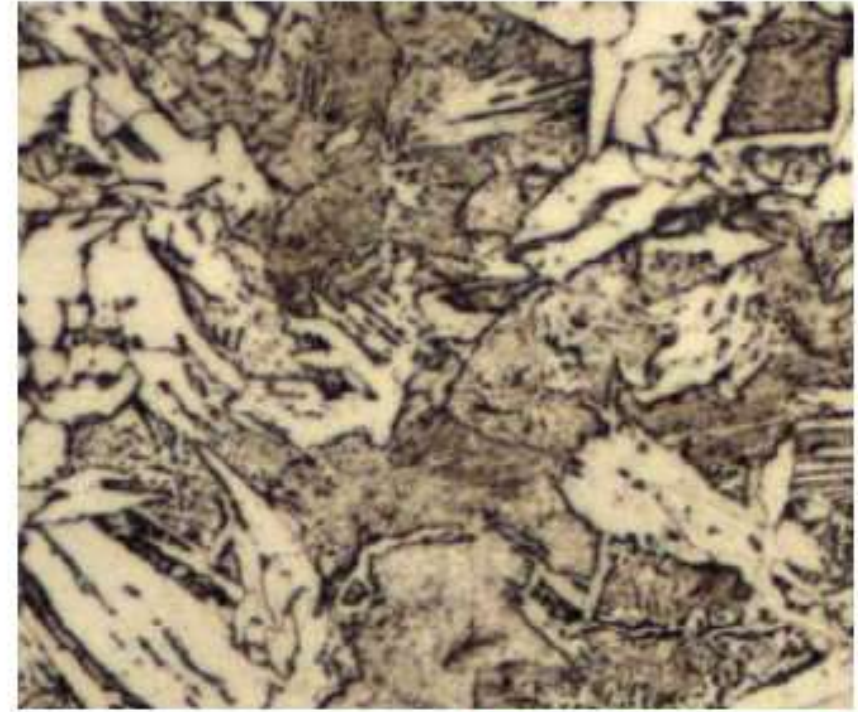
AIMS OF NORMALIZING

- To produce a harder and stronger steel than full annealing
- To improve machinability
- To modify and/or refine the grain structure
- To obtain a relatively good ductility without reducing the hardness and strength
- Improve dimensional stability
- Produce a homogeneous microstructure
- Reduce banding
- Provide a more consistent response when hardening or case hardening

EFFECT OF SOAKING TIME ON THE MICROSTRUCTURE:



(a)



(b)

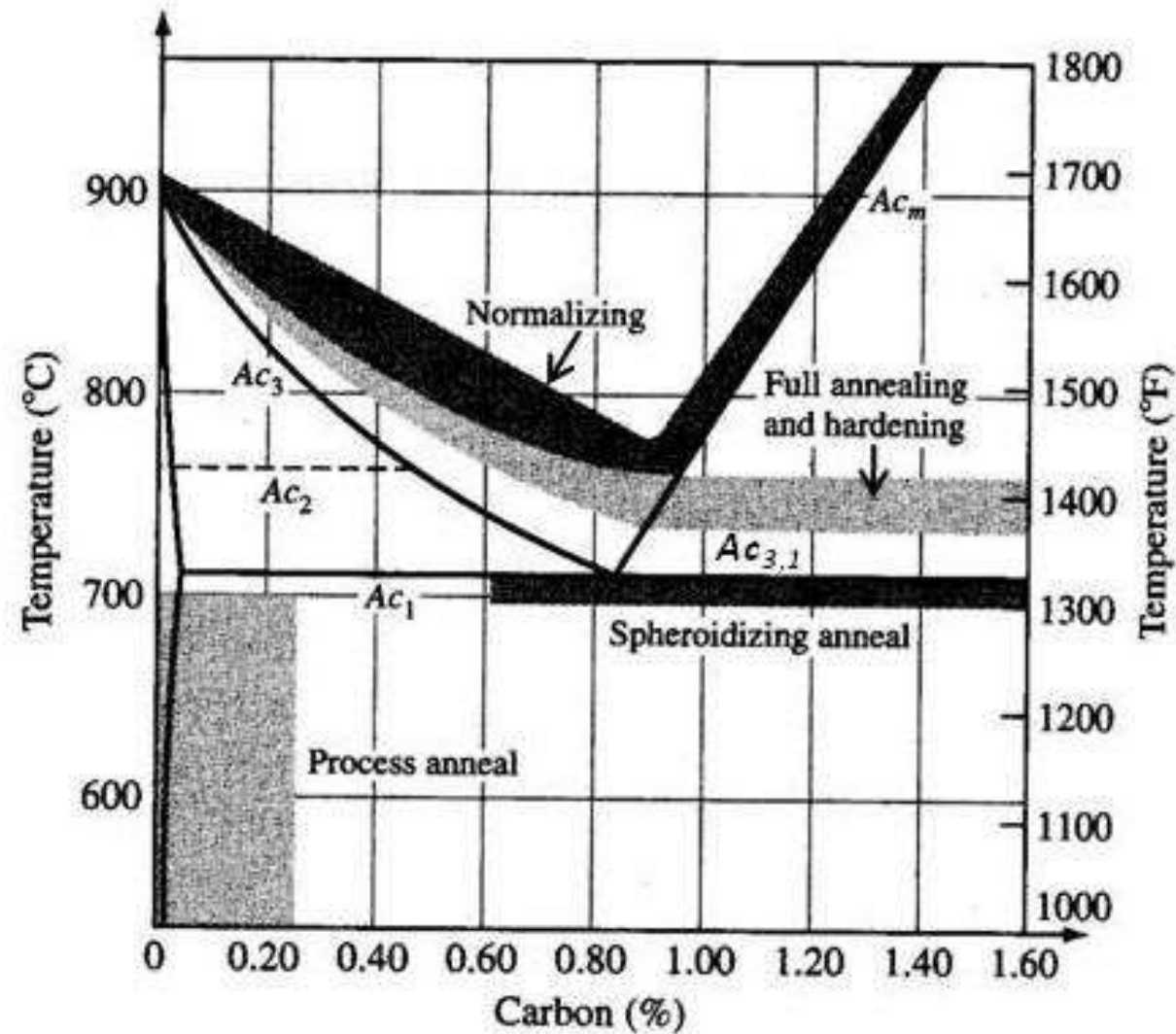
50 μm

Acicular ferrite in mixture with polygonal ferrite and pearlite, normalized at (a) 960°C, 10 minutes soaking time (b) 960°C, 40 minutes soaking time.

COMPARISON OF ANNEALING AND NORMALIZING

- The metal is heated to a higher temperature and then removed from the furnace for air cooling in normalizing rather than furnace cooling.
- In normalizing, the cooling rate is slower than that of a quench-and temper operation but faster than that used in annealing. As a result of this intermediate cooling rate, the parts will possess a hardness and strength somewhat greater than if annealed.
- Fully annealed parts are uniform in softness (and machinability) throughout the entire part; since the entire part is exposed to the controlled furnace cooling. In the case of the normalized part, depending on the part geometry, the cooling is non-uniform resulting in non-uniform material properties across the part.
- Internal stresses are more in normalizing as compared to annealing.
- Grain size obtained in normalizing is finer than in annealing.
- Normalizing is a cheaper and less time-consuming process.

Comparison of temperature ranges in annealing and normalizing



Temperature Range for Heat Treatment of Carbon Steels